# Can a mix be safer than either fund?

**Stock fund**

**20% volatility**

**Bond fund**

**10% volatility**

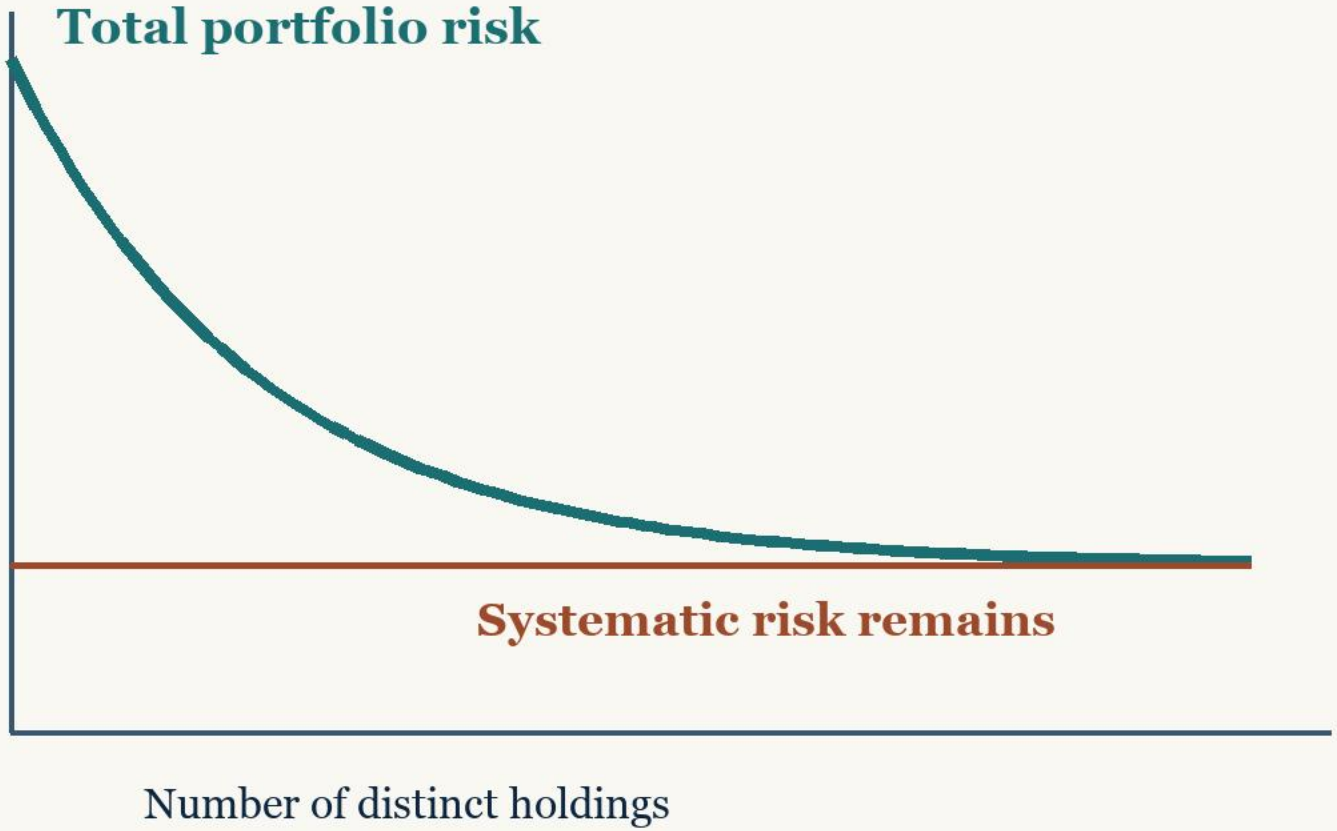


## A stock–bond mix

Predict its volatility

Diversification depends on how returns move together.

# Diversification removes firm-specific risk




## Firm-specific

Recall, lawsuit, management change

## Systematic

Recession, inflation, interest rates

A broad portfolio still faces recessions, inflation and rate changes.

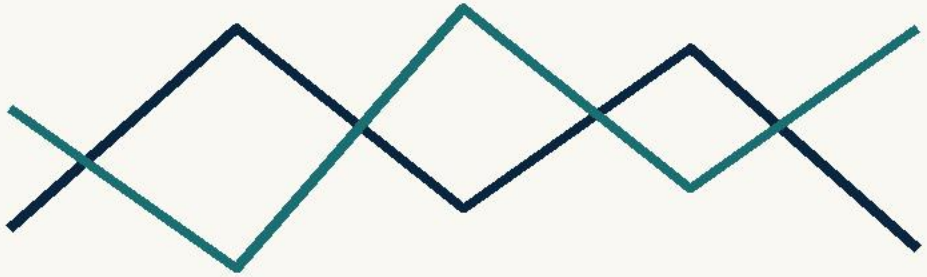
# Different return patterns create the benefit

## Same direction



Stock fund Second fund

## Different patterns



Stock fund Second fund

Schematic return patterns across the same periods

More holdings help when their risks do not all move together.

# Start with a distribution of possible returns

| State | Probability | Stock return | Bond return |
|---|---|---|---|
| Expansion | 20% | 25% | 4% |
| Stable growth | 50% | 10% | 6% |
| Contraction | 30% | −10% | 2% |

Probabilities total 100%. Each row pairs returns in the same economic state.

These are possible outcomes with known probabilities, not historical observations.

# Expected return is the probability-weighted mean

$$\mu = E(R) = \Sigma\, p(i)R(i)$$

p(i) = state probability; R(i) = return in that state; μ = expected return

### Weight each stock return

$$0.20 \times 25\% + 0.50 \times 10\% + 0.30 \times (-10\%)$$

### Add the contributions

$$5\% + 5\% - 3\% = 7.00\%$$

### Repeat for bonds

$$0.20 \times 4\% + 0.50 \times 6\% + 0.30 \times 2\% = 4.40\%$$

A 7% expected return is a long-run average across scenarios, not a promise.

# Variance measures distance from the mean

$$\sigma^2 = \Sigma\, p(i)(R(i) - \mu)^2;\ \sigma = \sqrt{\sigma^2}$$

| Stock state | Deviation from 7% | Weighted squared deviation |
|---|---|---|
| Expansion | $0.25 - 0.07 = 0.18$ | $0.20 \times 0.18^2 = 0.00648$ |
| Stable growth | $0.10 - 0.07 = 0.03$ | $0.50 \times 0.03^2 = 0.00045$ |
| Contraction | $-0.10 - 0.07 = -0.17$ | $0.30 \times (-0.17)^2 = 0.00867$ |

$\sigma^2 = 0.01560$
$\sigma = 12.49\%$

Square deviations so gains and losses cannot cancel. SD returns to return units.

# Build the statistics with cell references

F5: =(C5-$C$9)^2*B5

| Cell / range | Stock calculation | Bond calculation |
|---|---|---|
| C9 / D9 | =SUMPRODUCT(B5:B7,C5:C7) | =SUMPRODUCT(B5:B7,D5:D7) |
| F5 / G5 | =(C5-$C$9)^2*B5 | =(D5-$D$9)^2*B5 |
| F9 / G9 | =SUM(F5:F7) | =SUM(G5:G7) |
| B18 / C18 | =SQRT(F9) | =SQRT(G9) |

Fill F5:G5 through row 7. Results: stock μ 7%, σ 12.49%; bond μ 4.4%, σ 1.744%.

$ locks the mean while each state return and probability move down a row.

# Covariance compares deviations in the same state

$$c = \mathrm{Cov}(S,B) = \Sigma\, p(i)(RS(i) - \mu S)(RB(i) - \mu B)$$

S = stock; B = bond; c = covariance; μ = each fund's own mean

| | | | | |
|---|---|---|---|---|
| **Stock: Above** | × | **Bond: Above** | → | **Positive** |
| **Stock: Below** | × | **Bond: Below** | → | **Positive** |
| **Stock: Above** | × | **Bond: Below** | → | **Negative** |

Both above or both below their means add positive covariance.

# Calculate the covariance one state at a time

| State | p × stock deviation × bond deviation | Contribution |
|---|---|---|
| Expansion | 0.20 × 0.18 × (−0.004) | −0.000144 |
| Stable growth | 0.50 × 0.03 × 0.016 | 0.000240 |
| Contraction | 0.30 × (−0.17) × (−0.024) | 0.001224 |

c = 0.001320
Means: stock 7%; bond 4.4%

Weight the paired products by their probabilities, then add them.

# Correlation puts covariance on a common scale

$$\rho = c / (\sigma S \times \sigma B)$$

ρ = correlation, from −1 to +1; σS and σB = fund standard deviations

0.001320 ÷ (0.124900 × 0.017436)

**ρ = +0.606**




Positive correlation below +1 still permits diversification.

# Your turn: compare two equally likely states

| State | Probability | Stock return | Bond return |
|---|---|---|---|
| Strong economy | 50% | 20% | 2% |
| Weak economy | 50% | 0% | 6% |

Find both means and SDs. Then calculate covariance and correlation. Predict the sign first.

Use each asset's own mean and match returns from the same state.

# Opposite deviations create negative correlation

| Measure | Stock fund | Bond fund |
|---|---|---|
| Mean | 10% | 4% |
| Variance | 0.0100 | 0.0004 |
| Standard deviation | 10% | 2% |

$$c = 0.5(0.10)(-0.02) + 0.5(-0.10)(0.02) = -0.002$$

$$\rho = -0.002 / (0.10 \times 0.02) = -1$$

Here, one fund is above its mean exactly when the other is below.

# Historical samples need sample statistics

## Known scenario probabilities

Weight returns by probabilities

Variance = $\Sigma\, p(i)(R(i) - \mu)^2$

Use SUMPRODUCT and SQRT

## Observed historical sample

Each observation is one data point

Sample variance divides by $n - 1$

Use VAR.S and STDEV.S

Q4 asks about a sample of 50 annual observations: use the .S functions.

# One Excel function estimates sample volatility

| Cell / range | Input or formula | Meaning / result |
|---|---|---|
| B5:B8 | 8%, −4%, 12%, 4% | Four hypothetical annual returns |
| B10 | =AVERAGE(B5:B8) | 5.00% mean |
| B11 | =VAR.S(B5:B8) | 0.00466667 variance |
| B12 | =STDEV.S(B5:B8) | 6.831% standard deviation |

Squared deviations total 0.014. Divide by 4 − 1, then take the square root.

Use one full, numeric return range. Do not include years or mix asset columns.

# Carry one stock–bond example through the model

| Annual input | Stock fund | Bond fund |
|---|---|---|
| Expected return | 12% | 6% |
| Standard deviation | 20% | 10% |
| Variance | 0.0400 | 0.0100 |

**Correlation: 0.20**

**Risk-free rate: 3%**

## Base mix: 60% stocks + 40% bonds

Keep expected returns, standard deviations and correlation consistent.

# Portfolio return weights the two fund means

$$E(R_P) = w_S\mu_S + w_B\mu_B;\ w_S + w_B = 1$$

w = portfolio share; μ = expected fund return; P = combined risky portfolio




**0.60 × 12% = 7.20%**

**0.40 × 6% = 2.40%**

**7.20% + 2.40% = 9.60%**

60% stocks and 40% bonds produce 9.60% expected return.

# Portfolio variance has three terms

$$\sigma P^2 = wS^2\sigma S^2 + wB^2\sigma B^2 + 2wSwB\ c$$

$c = \rho\sigma S\sigma B = 0.20 \times 0.20 \times 0.10 = 0.004$

**Stock term** $0.60^2 \times 0.20^2 = 0.01440$

**Bond term** $0.40^2 \times 0.10^2 = 0.00160$

**Co-movement term** $2 \times 0.60 \times 0.40 \times 0.004 = 0.00192$

**Add: 0.01440 + 0.00160 + 0.00192 = 0.01792**

$\sigma P = \sqrt{0.01792} = 13.39\%$, below the 16% weighted average of fund SDs.

# Your turn: increase the bond share

| Given | Value |
|---|---|
| Stock expected return / SD | 12% / 20% |
| Bond expected return / SD | 6% / 10% |
| Correlation | 0.20 |
| New weights | 20% stocks / 80% bonds |

Calculate expected return, covariance, portfolio variance and SD. Can this mix be less volatile than bonds alone?

Show all three variance terms before taking the square root.

# A small stock share can reduce bond-only risk

$$\sigma P^2 = 0.20^2(0.20^2) + 0.80^2(0.10^2) + 2(0.20)(0.80)(0.004)$$

**Expected return**

$0.20 \times 12\% + 0.80 \times 6\% = 7.20\%$

**Portfolio variance**

$0.00160 + 0.00640 + 0.00128 = 0.00928$

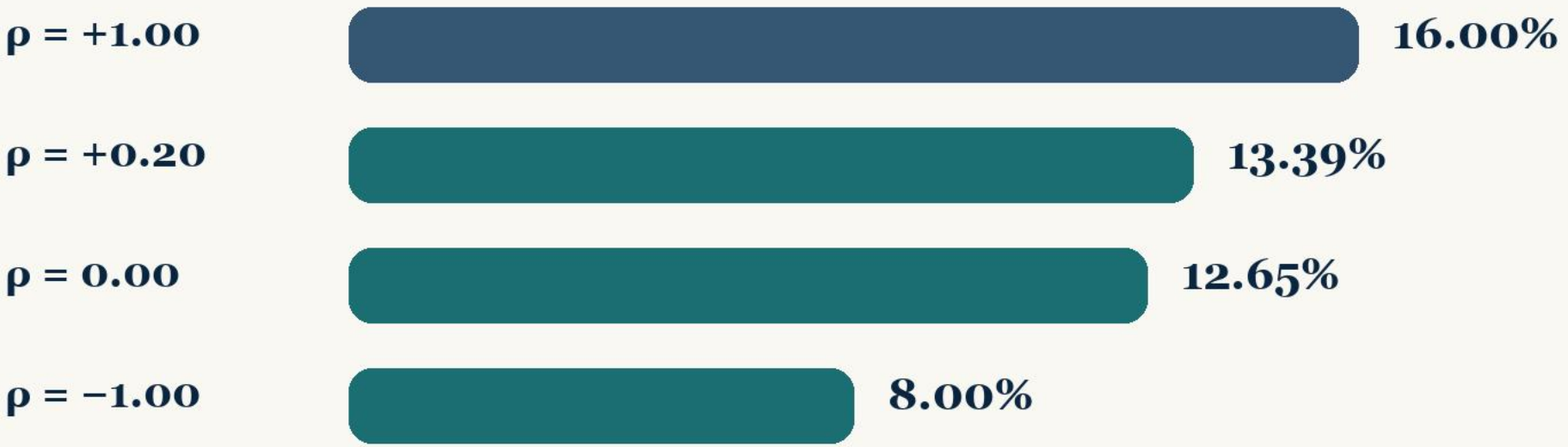
**Portfolio standard deviation**

$\sqrt{0.00928} = 9.633\% < 10\%$ bond-only SD

Imperfect co-movement can make the mix safer than either fund alone.

# Lower correlation lowers risk at fixed weights

$$\sigma P^2 = wS^2\sigma S^2 + wB^2\sigma B^2 + 2wSwB\rho\sigma S\sigma B$$




ρ = −1 can produce zero risk only at special offsetting weights.

Expected return stays 9.60%; only the co-movement assumption changes.

# Vary weights without moving fixed inputs

| Cell / range | Input or formula | Purpose |
|---|---|---|
| B5:C5; B6:C6 | 12%, 6%; 20%, 10% | Means; SDs |
| A9; A17:A27 | 0.20; 0%, 10%, ..., 100% | Correlation; stock weights |
| B17 | =A17*$B$5+(1-A17)*$C$5 | Portfolio mean |
| C17 | See formula below | Portfolio SD |

C17: =SQRT((A17*$B$6)^2+((1-A17)*$C$6)^2
+2*A17*(1-A17)*$A$9*$B$6*$C$6)

Fill down: A17 changes to A18; the means, SDs and correlation stay locked.

# Minimum variance finds the least volatile mix

$$w_{S,MV} = (\sigma_B^2 - c) / (\sigma_S^2 + \sigma_B^2 - 2c)$$

MV = minimum variance; $w_{B,MV} = 1 - w_{S,MV}$; c = covariance

### Use the running-example risk inputs

$\sigma_S^2 = 0.04$; $\sigma_B^2 = 0.01$; $c = 0.004$

### Calculate the stock share

$(0.01 - 0.004)/(0.04 + 0.01 - 0.008) = 14.29\%$

### Calculate the resulting risk

85.71% bonds; expected return 6.857%; SD 9.562%

Minimum variance minimizes risk. It does not maximize the Sharpe ratio.

# The efficient frontier is the upper boundary




Illustrative long-only two-fund opportunity set

## Upper branch

Efficient risky portfolios

## Leftmost point

Global minimum variance

## Many-asset inputs

N means, N variances,
N(N − 1)/2 covariances

At the same risk, prefer the higher expected return under mean–variance assumptions.

# Which candidate has the best reward per risk?

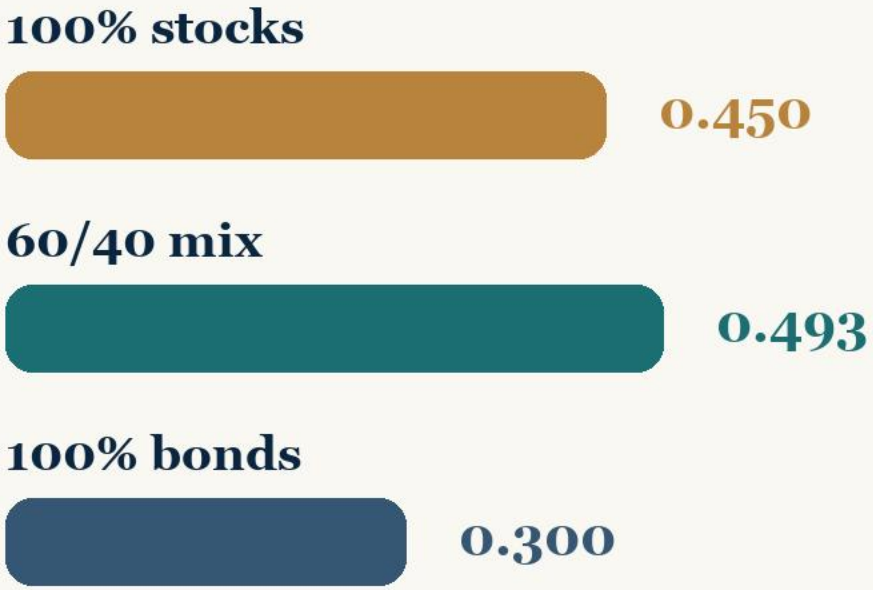
$$\text{Sharpe} = [E(RP) - Rf] / \sigma P$$

| Candidate | Expected return | Standard deviation |
|---|---|---|
| 100% stocks | 12.00% | 20.00% |
| 60% stocks / 40% bonds | 9.60% | 13.3866% |
| 100% bonds | 6.00% | 10.00% |

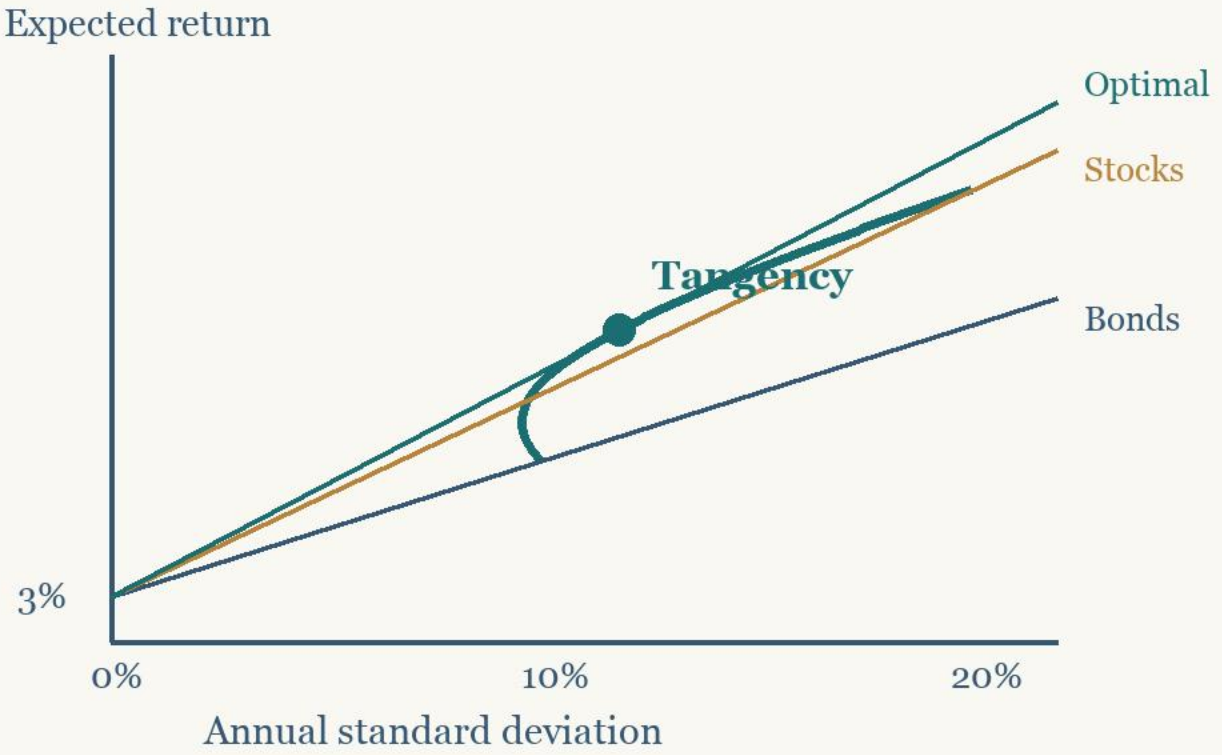
Rf = 3%. Calculate and rank the three Sharpe ratios. Does the best candidate prove a global optimum?

Sharpe compares excess expected return per unit of total volatility.

# 60/40 wins the comparison, but is not optimal




## Steeper CAL = higher Sharpe




Testing three candidates does not search every possible stock–bond mix.

# Tangency weights maximize the Sharpe ratio

$$w_B^* = (e_B\sigma_S^2 - e_S\, c) / [e_B\sigma_S^2 + e_S\sigma_B^2 - (e_B + e_S)c]$$

$e_B = \mu_B - R_f$; $e_S = \mu_S - R_f$; $c = \rho\sigma_S\sigma_B$; $w_S^* = 1 - w_B^*$

## Running example

$e_S = 12\% - 3\% = 9\%$

$e_B = 6\% - 3\% = 3\%$

$c = 0.004$

$\sigma_S^2 = 0.04$; $\sigma_B^2 = 0.01$

Bond numerator: $0.03 \times 0.04 - 0.09 \times 0.004 = 0.00084$

**Denominator: 0.00162; bond weight = 51.85%**

The optimal risky mix uses excess returns, variances and covariance.

# Construct the tangency weights in Excel

| Cell / range | Input or formula | Meaning |
|---|---|---|
| B5:C5 | 3%, 9% | Bond / stock excess returns |
| B6:C6; B9 | 10%, 20%; 0.004 | Bond / stock SD; covariance |
| B17 | Numerator ÷ denominator | Optimal bond weight |
| B18 | =1-B17 | Optimal stock weight |

B17: =(B5*C6^2-C5*B9)/
(B5*C6^2+C5*B6^2-(B5+C5)*B9)

51.85% bonds + 48.15% stocks: expected return 8.889%, SD 11.815%.

# Total returns must become excess returns first

## Q6: excess returns supplied

Use the given excess returns directly

Bond inputs are in column B

Weights belong in B17:B18

## Q7: total returns supplied

Subtract the Treasury-bill return

Stock excess return: B20

Stock weight: C20; bond: C21

Use the same method, but read each question’s labels and column order.

# Choose the risky mix, then choose its size

**Optimal risky mix**

**48.15% stocks**
**51.85% bonds**

**Choose y = 60% in the risky mix**




$E(RC) = yE(RP) + (1 - y)Rf; \sigma C = y\sigma P$

CAL = bills + a risky portfolio. CML uses the market portfolio.

Risk preference changes the allocation to the risky mix and bills.

# Diversification is powerful, but has limits

| | |
|---|---|
| **Estimation** | Expected returns and correlations change; test assumptions. |
| **Pooling and sharing** | Pooling adds exposures. Sharing spreads a fixed risk among investors. |
| **Time horizon** | More years do not guarantee smaller cumulative losses. |

Q1–Q2: measure / Q3: combine / Q4: sample / Q5: vary / Q6–Q7: optimize

Before trusting a portfolio, check inputs, formulas, constraints and economic meaning.